

Experiment 2

Problem Identification and Feasibility: Software Design Principles

Loan Eligibility Prediction System

MEMBER 1	DIKSHA VERMA (2400290130074)
MEMBER 2	DIPTI GUPTA (2400290130075)
MEMBER 3	KASHIKA MAURYA (2400290130100)
REPO LINK	https://github.com/Dipti18-Guptq/Loan-Eligibility-Prediction-System

Experiment 3

Software Design Principles – Requirement Elicitation

Aim

To identify stakeholders and perform requirement elicitation for the Loan Eligibility Prediction System using suitable techniques such as interviews, questionnaires, observation, and user personas; prepare user stories and develop a Requirement Gathering Report.

Objectives

- Identify key stakeholders of the Loan Eligibility Prediction System.
- Apply requirement elicitation techniques.
- Gather functional and non-functional requirements.
- Prepare user stories with acceptance criteria.
- Document the Requirement Gathering Report.

Introduction

Requirement elicitation is the process of discovering stakeholder needs and expectations. The quality of software depends on how accurately user requirements are captured. Common elicitation techniques include interviews, questionnaires, observation, workshops and user personas.

Selected Project

Project Title: Loan Eligibility Prediction System

Domain: Banking and Financial Services

Stakeholder Identification

Stakeholder	Role	Requirements
Applicant	Provides personal, employment and financial information	Simple input, quick response and clear prediction
Bank Employee	Reviews preliminary eligibility information	Understandable result and relevant applicant details
System Administrator	Maintains system and model resources	Easy maintenance, monitoring and controlled access
Project Team	Designs, develops and tests the system	Clear requirements, documentation and reliable implementation

Requirement Elicitation Techniques

Technique	Purpose	Outcome
Interview	Discuss loan assessment needs with stakeholders	Detailed functional requirements
Questionnaire	Collect feedback from multiple potential users	User expectations and preferences
Observation	Observe how users provide	Usability and workflow

	information and understand results	requirements
User Persona	Model representative users	User-centred requirements

Sample Requirement Gathering

Interview Findings

- Applicants require a simple and quick way to enter their information.
- Users want the prediction result to be easy to understand.
- Users should be informed when required information is missing or invalid.
- The system should clearly distinguish a prediction from final loan approval.

Observation Findings

- Users prefer simple navigation and clearly labelled input fields.
- Frequently required applicant details should be easy to locate.
- The prediction result should be visible without unnecessary steps.

Questionnaire Findings

- Users expect quick prediction results.
- Users consider privacy and security of financial information important.
- Users prefer simple explanations instead of technical machine-learning terminology.

User Personas

Persona	Age	Goal	Pain Point
First-time Applicant	22	Know whether the application appears eligible	Unfamiliar with loan criteria
Working Professional	30	Get a quick preliminary eligibility indication	Time-consuming manual assessment
Self-employed Applicant	40	Understand eligibility using financial details	Complex financial information

User Stories

ID	User Story	Acceptance Criteria
US1	As an applicant, I want to enter my personal and financial details so that the system can evaluate my application.	Required fields are accepted and valid input passes validation.
US2	As an applicant, I want the system to validate my information so that I can correct missing or invalid details.	Invalid or missing values produce clear validation messages.
US3	As an applicant, I want to receive a preliminary eligibility prediction so that I can understand the likely result of my application.	A valid submission produces a clearly displayed prediction.
US4	As an applicant, I want the prediction to be explained in	The explanation is clear and is presented as supplementary

	simple language so that I can understand the result.	information.
US5	As a bank employee, I want to review applicant information and the prediction so that I can use it as preliminary decision-support information.	Relevant applicant details and prediction are displayed correctly.

Requirement Gathering Report

The elicitation process identified applicant information entry, validation, preprocessing, eligibility prediction, result display, explanation, security and usability as the primary requirements. Stakeholder feedback emphasised ease of use, fast response time, understandable results, reliability and secure handling of applicant information.

Observation

Using multiple elicitation techniques improves requirement completeness and reduces ambiguity. Combining stakeholder discussions, questionnaires, observation and user personas helps capture both functional and user-centred requirements.

Result

Stakeholders were identified, requirements were elicited using suitable techniques, user stories were prepared, and the Requirement Gathering Report for the Loan Eligibility Prediction System was successfully completed.